SSC Maharashtra Board — Science and Technology (Part II)

March 2026 — Model Solutions

Time: 2 Hours Max. Marks: 40

Q.1

Q.1 (A) Choose the correct alternative: [5 Marks]

(i) The causality behind the sudden changes was understood due to ... principle of Hugo de Vries.

(A) Appendix (B) Mutation (C) Transcription (D) Translocation

Hugo de Vries proposed the **Mutation theory** to explain sudden, inheritable changes in organisms. He observed these sudden changes (mutations) in the evening primrose plant (*Oenothera lamarckiana*).

Ans: (B) Mutation

(ii) After complete oxidation of a glucose molecule, ... number of ATP molecules are formed.

(A) 18 (B) 28 (C) 38 (D) 48

During aerobic respiration, one molecule of glucose ($C_6H_{12}O_6$) undergoes complete oxidation through glycolysis, Krebs cycle, and the electron transport chain, producing a total of **38 ATP molecules**.

Ans: (C) 38

(iii) Bat is included in class ...

(A) Amphibia (B) Reptilia (C) Aves (D) Mammalia

Although bats can fly, they are not birds. Bats are **mammals** because they possess mammary glands (give milk to young ones), have hair on their body, are warm-blooded, and give birth to live young ones.

Ans: (D) Mammalia

(iv) In Maharashtra the major hydroelectric power plant is located at ...

(A) Chandrapur (B) Koyana (C) Parali (D) Tarapur

The **Koyana Hydroelectric Project** is located in Satara district of Maharashtra. It is the largest completed hydroelectric project in Maharashtra, built on the Koyana River. (Chandrapur and Parali have thermal power plants; Tarapur has a nuclear power plant.)

Ans: (B) Koyana

(v) A first state to start a separate cyber crime unit in our country is . . .

(A) Madhya Pradesh (B) Maharashtra (C) Gujrat (D) Karnataka

Maharashtra was the first state in India to establish a separate cyber crime cell/unit to handle cyber-related offences.

Ans: (B) Maharashtra

Q.1 (B) Answer the following questions: [5 Marks]

(i) Fill in the box with proper words:

Asexual reproduction in Unicellular Organisms

Asexual reproduction in Unicellular Organisms

Binary fission | Multiple fission | Budding

The three types of asexual reproduction in unicellular organisms are Binary fission (e.g., Amoeba), Multiple fission (e.g., Plasmodium), and **Budding** (e.g., Yeast).

Budding

(ii) Which day is observed as World Biodiversity Day?

World Biodiversity Day is observed on **22nd May** every year.

It was proclaimed by the United Nations to increase understanding and awareness of biodiversity issues.

22nd May

(iii) Identify me, who am I?

"I am the first clone sheep born in Scotland."

Dolly

Dolly the sheep was the first mammal to be cloned from an adult somatic cell. She was born on 5th July 1996 at the Roslin Institute in Scotland.

(iv) Write the molecular formula of acetic acid.

Acetic acid (also known as ethanoic acid) has the molecular formula:

$$\boxed{CH_3COOH}$$

Its IUPAC name is Ethanoic acid, and its chemical formula can also be written as $C_2H_4O_2$.

(v) State whether the following statement is true or false:
"The chairman of National Disaster Management is Sarpanch."

$$\boxed{\text{False}}$$

The Chairman of the **National Disaster Management Authority (NDMA)** is the **Prime Minister of India**, not the Sarpanch. The Sarpanch is the head of the Gram Panchayat (village level).

Q.2

Q.2 (A) Give scientific reasons (any *two*): [4 Marks]

(i) The popularity of probiotic products is increasing.

1. Probiotic products contain **live beneficial microorganisms** (especially bacteria like *Lactobacillus* and *Bifidobacterium*) which are good for human health.
2. These beneficial bacteria help in **improving digestion** by maintaining a healthy balance of gut flora (microbiome) in the intestine.
3. They help in boosting the **immune system** and preventing harmful bacteria from growing in the intestine.
4. Probiotic products like curd, yogurt, and fermented foods help in absorption of nutrients, synthesis of vitamins, and prevention of diseases like diarrhoea and allergies.
5. Growing awareness about **preventive healthcare** and the benefits of good gut bacteria has increased the demand for probiotic products.

$\therefore$ The popularity of probiotic products is increasing.

(ii) Body temperature of reptiles is not constant.

1. Reptiles are **cold-blooded animals** (poikilotherms/ectotherms).
2. Their body temperature **changes with the surrounding environmental temperature**. They do not have an internal mechanism to regulate their body temperature.
3. They lack efficient **thermoregulatory mechanisms** like sweating or shivering that are present in warm-blooded animals (endotherms).

4. Reptiles depend on external sources of heat, such as sunlight, to warm their body. This is called **basking behaviour**.

5. Since the environmental temperature varies throughout the day and across seasons, the body temperature of reptiles also fluctuates accordingly.

$\therefore$ Body temperature of reptiles is not constant.

(iii) Nowadays farmers are opting for organic farming.

1. **Chemical fertilizers and pesticides** used in conventional farming cause soil degradation, water pollution, and harm to beneficial organisms.

2. Organic farming uses **natural fertilizers** (compost, vermicompost, green manure) and **bio-pesticides**, which are eco-friendly and do not contaminate soil and water.

3. Food produced through organic farming is **free from harmful chemical residues**, making it safer and healthier for consumption.

4. Organic farming helps maintain **soil fertility** and biodiversity in the long run.

5. There is growing **consumer demand** for organic food products due to increasing health awareness, which gives better market prices to farmers.

6. Organic farming is **sustainable** and helps in conservation of natural resources.

$\therefore$ Nowadays farmers are opting for organic farming.

Q.2 (B) Answer the following questions (any *three*): [6 Marks]

(i) What is meant by carbon dating? Give any *one* use of this technique.

Carbon dating (Radiocarbon dating):
Carbon dating is a scientific technique used to determine the **age of fossils, archaeological artefacts, and other objects** that contain organic material.

- All living organisms contain the radioactive isotope **Carbon-14 (C-14)** along with the stable isotope Carbon-12 (C-12).
- When an organism dies, it stops absorbing carbon. The C-14 in the dead organism begins to **decay at a known rate** (half-life of approximately 5,730 years).
- By measuring the **ratio of C-14 to C-12** remaining in a fossil or artefact, scientists can calculate how long ago the organism died.

Use of carbon dating:
Carbon dating is used to determine the **age of fossils** found in excavations, helping scientists understand the evolutionary history and timeline of organisms that lived in the past.

(ii) Complete the table:

Sr. No.	Protein	Name of Organ
1	$\boxed{\text{Ossein}}$	Bones
2	Insulin	$\boxed{\text{Pancreas}}$

Explanation:

- **Ossein** is the structural protein found in bones. It is the organic component of bone that gives it flexibility and tensile strength.
- **Insulin** is a protein hormone produced by the **beta cells of the Islets of Langerhans** in the **pancreas**. It regulates blood sugar (glucose) levels in the body.

(iii) Read the paragraph and answer the questions:

"Paddy is cultivated on large scale in various states of South India. Paddy fields are frequently attacked by grasshoppers. Similarly, frogs are also present in large number in the mud of paddy fields, to feed upon grasshoppers and snakes are also present therein to feed upon their favourite food-frogs. However, if frog population declines all of a sudden,"

(a) What will be the effect on paddy crop?
If the frog population declines suddenly:

- Frogs are the natural predators of grasshoppers. Without frogs, the **grasshopper population will increase** uncontrolled.
- The increased number of grasshoppers will cause **severe damage to the paddy crop** by feeding on it excessively.
- This will lead to **crop loss and reduced yield**.

$$\boxed{\text{The paddy crop will be severely damaged due to increase in grasshoppers.}}$$

(b) Number of which components will increase?
If frog population declines:

- The number of **grasshoppers** will increase (as their predator, frog, is absent).
- The number of **snakes may decrease** (as their food source, frogs, is reduced).

$$\boxed{\text{Grasshoppers (and other insect pests) will increase.}}$$

(iv) Write any *two* advantages of hydroelectric power generation.

1. **Renewable and clean energy source:** Hydroelectric power is generated from flowing water, which is a renewable resource. It does not produce air pollution, greenhouse gases, or toxic waste during electricity generation, unlike thermal power plants.

2. **Low operational cost:** Once a hydroelectric dam is constructed, the running cost of generating electricity is very low, as the fuel (water) is free and naturally available. The lifespan of hydroelectric power plants is also very long (50–100 years).

Additional advantages:

- Dams also help in **flood control**, irrigation, and water supply.
- Hydroelectric power generation is **reliable** and can be quickly started and stopped according to demand.

(v) The following are the pictures of some disasters. How will be your post disaster management in case you face any of those disasters?

The pictures show: (A) **Gas leak / Toxic chemical disaster** and (B) **Flood / Earthquake (building collapse)**.

Post-disaster management for Gas leak / Toxic chemical disaster:

1. Immediately **evacuate the affected area** and move to a safe location upwind.
2. **Cover nose and mouth** with a wet cloth to avoid inhaling toxic gases.
3. **Call emergency services** (fire brigade, ambulance, and disaster management authorities) immediately.
4. Provide **first aid** to affected persons (move to fresh air, give oxygen if available).
5. Do not use any electrical switches or open flames near the leak area.

Post-disaster management for Flood / Building collapse:

1. **Search and rescue** trapped or injured persons with the help of trained personnel.
2. Provide **first aid and medical help** to the injured.
3. Arrange for **temporary shelters, food, and clean drinking water** for affected families.
4. **Avoid entering damaged structures** until they are declared safe by authorities.
5. Report the disaster to the **local administration and NDMA** for further relief operations.

Q.3 Answer the following questions (any *five*): [15 Marks]

(i) Explain Lamarckism with two examples.

Lamarckism (Theory of Inheritance of Acquired Characters) was proposed by **Jean-Baptiste Lamarck** in 1809.

Main principles of Lamarckism:

1. **Internal vital force:** Every organism has an internal vital force that tends to increase its body size up to a certain limit.
2. **Effect of environment and new needs:** Changes in the environment lead to changes in the needs of organisms.
3. **Use and disuse of organs:** If an organ is used constantly, it becomes well-developed (**use**). If an organ is not used, it gradually degenerates and may eventually disappear (**disuse**).
4. **Inheritance of acquired characters:** The characters (modifications) acquired by an organism during its lifetime due to the use or disuse of organs can be **passed on to the next generation**.

Examples:
1. Long neck of the giraffe:

- According to Lamarck, the ancestors of giraffes had short necks and short forelimbs.
- Due to scarcity of food on the ground, they had to **stretch their necks to reach leaves on tall trees**.
- Over generations of continuous stretching, their necks and forelimbs became progressively longer.
- This acquired character was inherited by successive generations, resulting in the modern long-necked giraffe.

2. Webbed feet of aquatic birds:

- Aquatic birds like ducks and swans were originally land birds.
- When they started living near water bodies, they had to **spread their toes to swim**, stretching the skin between the toes.
- Over many generations, this continuous stretching led to the development of **webbed feet**.
- This acquired character was passed on to the offspring.

(ii) Explain the theory of Lamarckism with the help of one example — OR — Write the type of cells in which mitosis occurs. Write four stages of karyokinesis.

Type of cells in which mitosis occurs:

Mitosis occurs in **somatic cells (body cells)** and **stem cells**. It is also called equational division because the daughter cells have the same number of chromosomes as the parent cell.

Four stages of Karyokinesis (nuclear division) in Mitosis:

1. **Prophase:**
   - Chromatin fibres condense and become visible as distinct chromosomes.
   - Each chromosome consists of two sister chromatids joined at the centromere.
   - Nuclear membrane and nucleolus start to disappear.
   - Centrioles move to opposite poles and form spindle fibres.
2. **Metaphase:**
   - Chromosomes arrange themselves along the **equatorial plate** (metaphase plate) of the cell.
   - Spindle fibres attach to the centromere of each chromosome.
   - Chromosomes are most visible and distinct at this stage.
3. **Anaphase:**
   - The centromere of each chromosome splits.
   - Sister chromatids **separate and move to opposite poles** of the cell, pulled by the spindle fibres.
   - Each chromatid is now called a daughter chromosome.
4. **Telophase:**
   - Daughter chromosomes reach the opposite poles and begin to **decondense**.
   - Nuclear membrane and nucleolus **reappear** around each set of chromosomes.
   - Two daughter nuclei are formed, each with the same number of chromosomes as the parent cell.

After karyokinesis, **cytokinesis** (division of cytoplasm) occurs, resulting in two genetically identical daughter cells.

(iii) Give one example of each of the threatened species given below:
(a) Endangered species (b) Rare species (c) Vulnerable species

(a) Endangered species:

Endangered species are those whose population has decreased to a critical level and are in danger of extinction.

Example: Lion-tailed Macaque (*Macaca silenus*)

Other examples: Snow Leopard, Indian Rhinoceros, Asiatic Lion.

(b) Rare species:

Rare species have a small population size and are found in limited geographical areas. They are not yet endangered but are at risk.

Example: Red Panda (*Ailurus fulgens*)

Other examples: Himalayan Brown Bear, Wild Asiatic Buffalo.

(c) Vulnerable species:

Vulnerable species are those whose population is decreasing rapidly and may become endangered if the underlying causes are not addressed.

Example: Bengal Tiger (*Panthera tigris*)

Other examples: Indian Elephant, Gangetic Dolphin, Nilgiri Tahr.

(iv) Name the fuel used in thermal power plant. What are the problems faced in thermal power generation?

Fuel used in thermal power plant:

The main fuels used in thermal power plants are **coal**, natural gas, and petroleum (oil). In India, **coal** is the most commonly used fuel for thermal power generation.

Coal (fossil fuel)

Problems faced in thermal power generation:

1. **Air pollution:** Burning of coal releases large amounts of smoke, ash, sulphur dioxide (SO_2), nitrogen oxides (NO_x), and carbon dioxide (CO_2) into the atmosphere, causing severe air pollution.
2. **Greenhouse effect and global warming:** The CO_2 released during combustion of fossil fuels contributes to the greenhouse effect, leading to global warming and climate change.
3. **Non-renewable source:** Coal and other fossil fuels are non-renewable resources. Their reserves are finite and will eventually get exhausted.
4. **Ash disposal problem:** The burning of coal produces large quantities of fly ash and bottom ash, which create disposal problems and contaminate soil and water.

5. **Acid rain:** The SO_2 and NO_x emissions combine with atmospheric moisture to form sulphuric acid and nitric acid, resulting in acid rain that harms crops, buildings, and aquatic life.

6. **Water pollution:** Thermal power plants require large quantities of water for cooling, and the discharge of heated water into rivers affects aquatic ecosystems (thermal pollution).

(v) Identify the type of body symmetry of the animals shown in the given pictures:

The pictures show: (A) Amoeba, (B) A butterfly/Starfish, (C) A goat.

(A)

Asymmetrical body

(B)

Radial symmetry

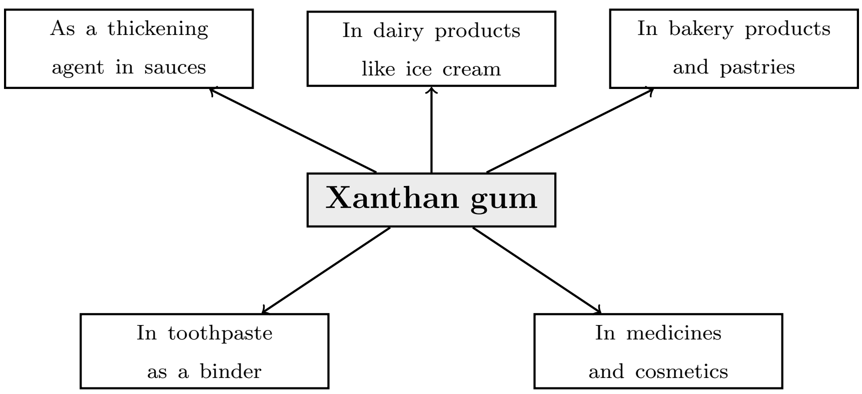
(C)

Bilateral symmetry

(vi) Complete the following conceptual picture with respect to uses of Xanthan gum:

Xanthan gum is a polysaccharide produced by the bacterium *Xanthomonas campestris* through fermentation. It is widely used as a food additive and thickening agent.

Uses of Xanthan gum:



Five important uses of Xanthan gum:

1. Used as a **thickening agent** in sauces, gravies, and dressings.
2. Used in **dairy products** like ice cream and yogurt to improve texture and prevent ice crystal formation.
3. Used in **bakery products** like bread, cakes, and pastries to improve moisture retention.
4. Used in **toothpaste** as a binder to maintain consistency.
5. Used in **medicines and cosmetics** as a stabilizer and emulsifier.

(vii) Which precautions will you take during spraying of pesticides?

The following precautions should be taken during spraying of pesticides:

1. **Wear protective clothing:** Always wear full-sleeved clothes, gloves, boots, goggles, and a face mask/respirator to avoid direct contact with pesticides.

2. **Cover nose and mouth:** Use a proper mask or a wet cloth to prevent inhalation of pesticide fumes and droplets.

3. **Spray in the direction of wind:** Always spray in the direction of the wind so that the pesticide spray does not blow back towards you.

4. **Wash hands and body:** After spraying, thoroughly wash hands, face, and other exposed body parts with soap and clean water. Change clothes immediately.

5. **Do not eat, drink, or smoke:** Avoid eating, drinking, or smoking during spraying to prevent accidental ingestion of pesticides.

6. **Keep children and animals away:** Ensure that children, pets, and livestock are kept away from the spraying area.

7. **Proper storage:** Store pesticides in original containers with labels, in a locked place away from food items and the reach of children.

8. **Follow instructions:** Read and follow the instructions on the pesticide label regarding dosage, dilution, and application method.

(viii) Write the names of any six materials in a first-aid box.

A first-aid box should contain the following essential materials:

1. **Sterile cotton** (absorbent cotton wool)

2. **Bandage** (roller bandage and triangular bandage)

3. **Antiseptic solution** (e.g., Dettol, Betadine, or Savlon)

4. **Adhesive plasters / Band-aids** (for covering small cuts and wounds)

5. **Scissors** (for cutting bandages, tape, and clothing)

6. **Thermometer** (to check body temperature)

Additional useful items include: pain-relief tablets (paracetamol), ORS packets, safety pins, tweezers, disposable gloves, burn ointment, and a torch.

Q.4 Answer the following questions (any *one*): [5 Marks]

(i) Social Health

(a) Write the names of any six factors affecting social health:
The following factors affect social health:

1. **Family environment:** A supportive and loving family environment contributes positively to social health.
2. **Education:** Access to quality education develops knowledge, skills, and the ability to interact meaningfully with society.
3. **Economic condition:** Financial stability provides access to healthcare, nutrition, and a better quality of life.
4. **Social equality:** Absence of discrimination based on caste, religion, gender, or economic status promotes social harmony.
5. **Hygiene and sanitation:** Clean surroundings and proper sanitation prevent diseases and promote community health.
6. **Addiction avoidance:** Staying away from alcohol, tobacco, and drug abuse is essential for maintaining social health.

Other factors include: government policies, cultural values, access to healthcare, peer group influence, and mental well-being.

(b) "Your brother studying in XII has developed the stress." — How will you manage this?
Managing stress in a student studying for XII board exams:

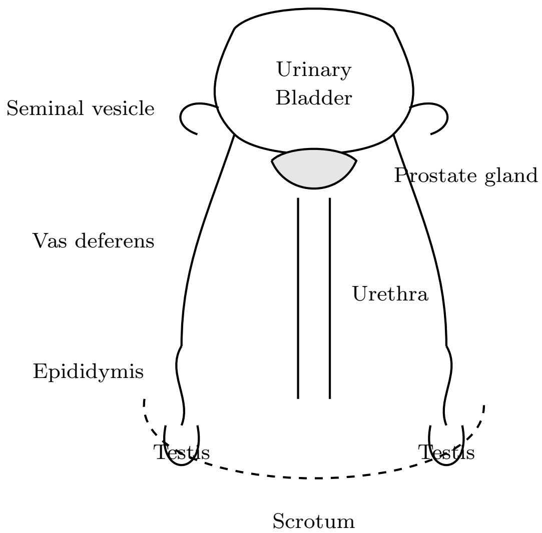
1. **Talk and listen:** Have a calm, supportive conversation with your brother. Listen to his concerns about studies, exams, or peer pressure without judging.
2. **Create a study schedule:** Help him create a **realistic and balanced timetable** that includes dedicated study hours, regular breaks, and leisure activities.
3. **Encourage physical activity:** Encourage him to engage in regular physical exercises like walking, yoga, or sports, which help reduce stress hormones and improve mood.
4. **Ensure proper nutrition and sleep:** Make sure he gets a **balanced diet** and at least 7–8 hours of sleep every night for optimal brain function.
5. **Positive motivation:** Remind him that **marks do not define success**. Encourage him to focus on understanding concepts rather than just memorizing.
6. **Seek professional help if needed:** If the stress is severe and persistent, consult a **counsellor or psychologist** who specializes in student mental health.

7. **Reduce screen time:** Limit unnecessary use of social media and electronic devices, especially before bedtime.

8. **Relaxation techniques:** Teach him relaxation techniques like **deep breathing, meditation, or listening to soothing music** to calm the mind.

(ii) Male Reproductive System

Observe the diagram given below and answer the questions:



(1) Write the name of the reproductive system in the above diagram. [1 Mark]

$\boxed{\text{Male Reproductive System}}$

(2) Name any *two* glands in the above diagram. $[\frac{1}{2} + \frac{1}{2} = 1$ **Mark]**

$\boxed{\text{1. Prostate gland} \qquad \text{2. Seminal vesicle}}$

The prostate gland and seminal vesicles are **accessory glands** that produce seminal fluid (semen), which nourishes, protects, and provides a transport medium for sperms.

(3) Mention any *one* duct in the above system. [1 Mark]

$\boxed{\text{Vas deferens (Sperm duct)}}$

The vas deferens is the duct that carries sperms from the epididymis to the urethra during ejaculation.

(4) What is the length of a sperm? [1 Mark]

$\boxed{\text{Length of a sperm} \approx 60\,\mu\text{m (micrometres)}}$

A human sperm cell is approximately 60 micrometres (0.06 mm) in length. It consists of a head, middle piece, and a long tail (flagellum) for movement.

(5) Which hormone is secreted by the testes? [1 Mark]

$\boxed{\text{Testosterone}}$

Testosterone is the primary male sex hormone (androgen) secreted by the **Leydig cells (interstitial cells)** of the testes. It is responsible for the development of male secondary sexual characters such as growth of facial hair, deepening of voice, and muscular development.

— **End of Solutions** —